Experiment 2

Conjugate Heat Transfer in a Double-Pipe Heat Exchanger

Aim

To study thermal interaction between hot and cold laminar streams separated by a conducting wall, including the influence of cold-side Reynolds number and wall-to-fluid heat-transfer ratio.

Simulation Parameters

Parameter	**Value**
Streamwise grid points	160
Cold-fluid cross-stream points	20
Wall cross-stream points	6
Hot-fluid cross-stream points	20
Cold-side Reynolds number	50, 100, 200
Hot-side Reynolds number	100
Solid/fluid heat-transfer ratio	1 and 100
Time step	1.25×10^{-3}
Iterations	3000

Numerical Method

A steady finite-volume convection–diffusion model is used for the two fluid passages and is coupled to heat conduction in the separating wall. Prescribed parabolic velocity profiles are used to focus the calculation on conjugate thermal transport.

Boundary Conditions

Hot and cold inlet temperatures are prescribed. Zero-gradient conditions are applied at the outlets, the external boundaries are treated as adiabatic, and temperature and heat flux are continuous across each fluid–solid interface.

Observations

Outlet temperatures, heat-transfer rate, effectiveness, temperature residual, relative energy imbalance, and temperature contours were considered for different cold-side Reynolds numbers and conductivity ratios.

Results

Cold-side Re	Solid/fluid ratio	Hot outlet (normalised)	Cold outlet (normalised)
50	1	0.98	0.025
100	1	0.97	0.035
200	1	0.98	0.020
100	100	0.96	0.040
200	100	0.965	0.030

Conclusion

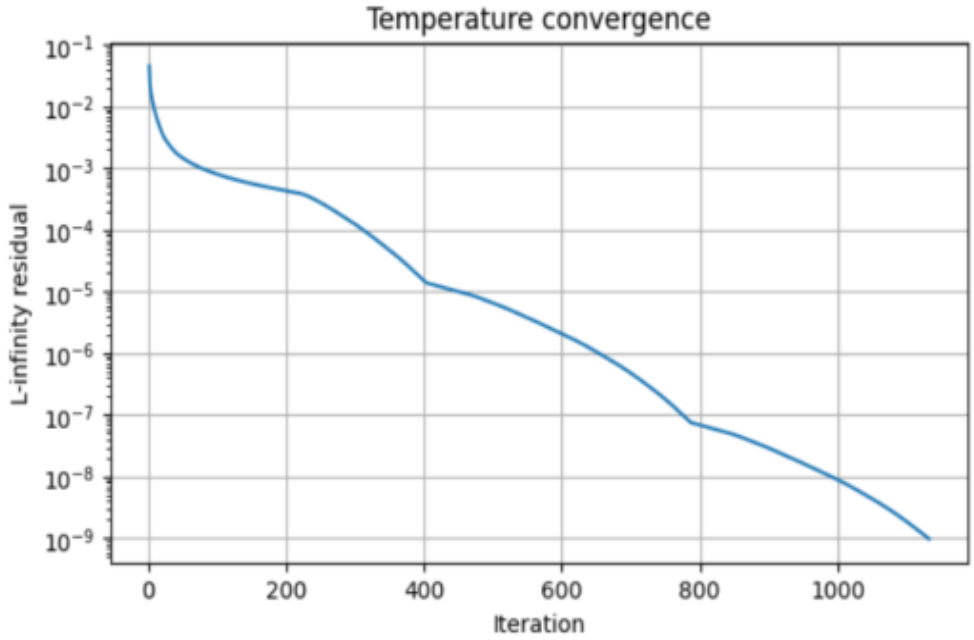
The temperature contours show heat moving from the hot passage toward the cold passage through the separating wall. Outlet temperatures settle as iterations proceed, while the residual decreases substantially. The results indicate that the flow Reynolds number and the relative thermal conductivity of the wall influence the predicted temperature field and outlet conditions.

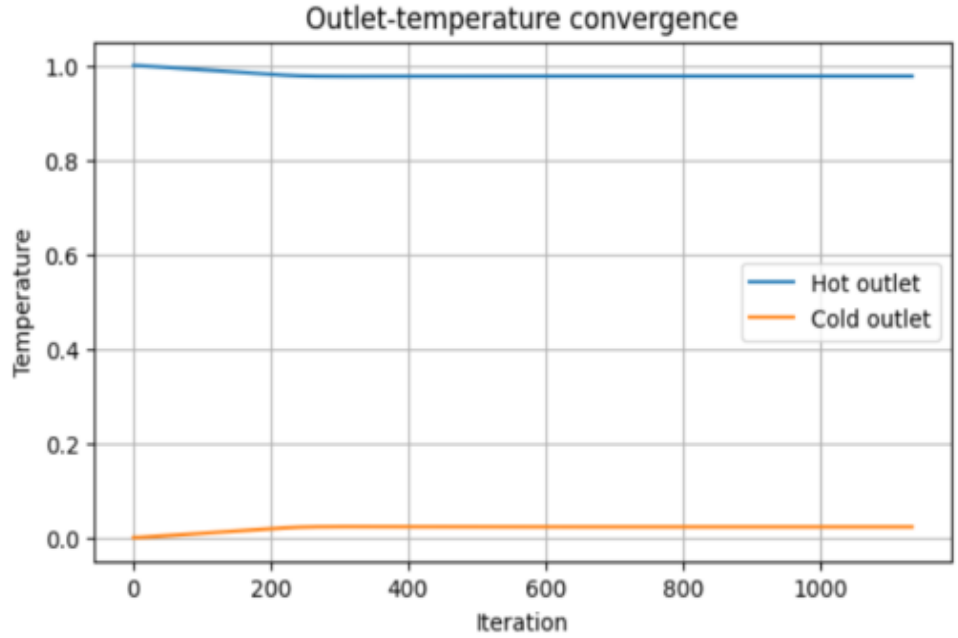
Original Simulation Figures

The following pages contain the plots and simulation outputs from the supplied report.

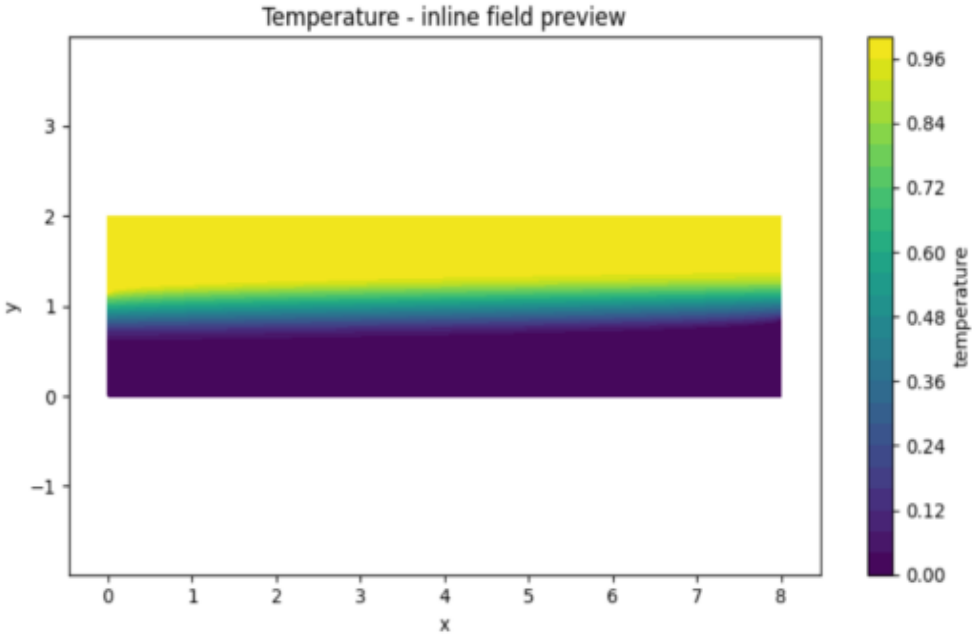
Original simulation output (source page 2)

For Counter flow system Raynold's number 50 and wall to fluid ratio 1

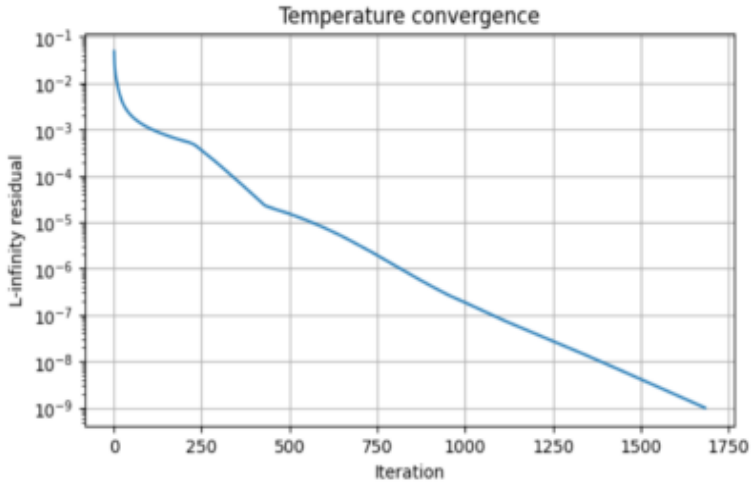




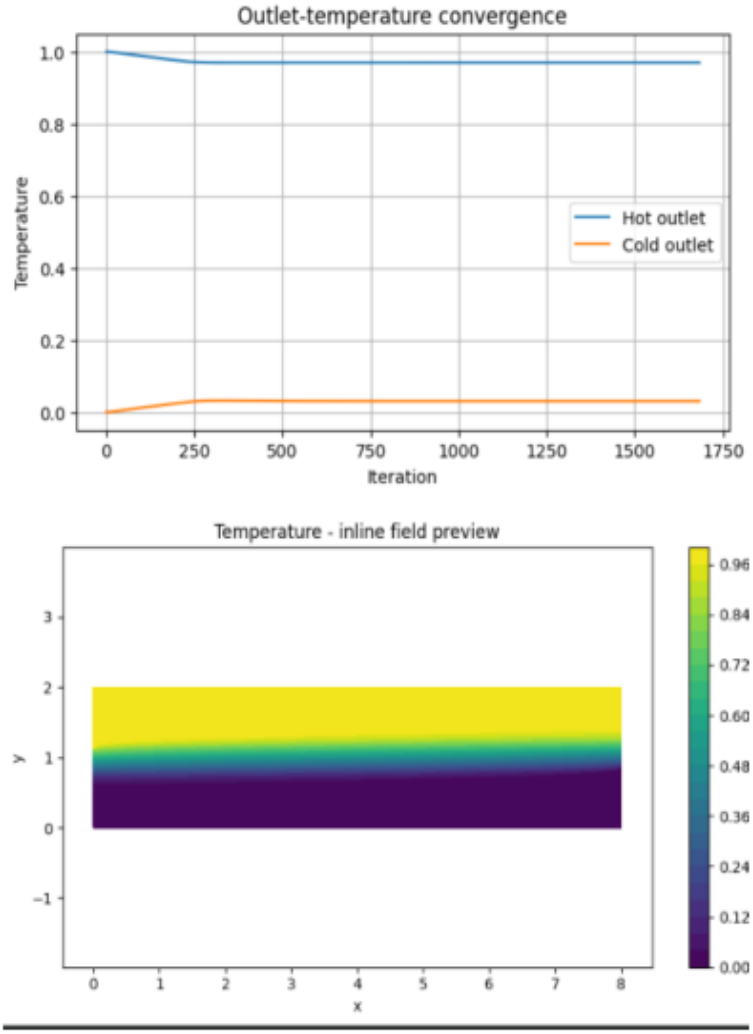
Original simulation output (source page 3)



For Counter flow system Raynold's number 100 and wall to fluid ratio 1

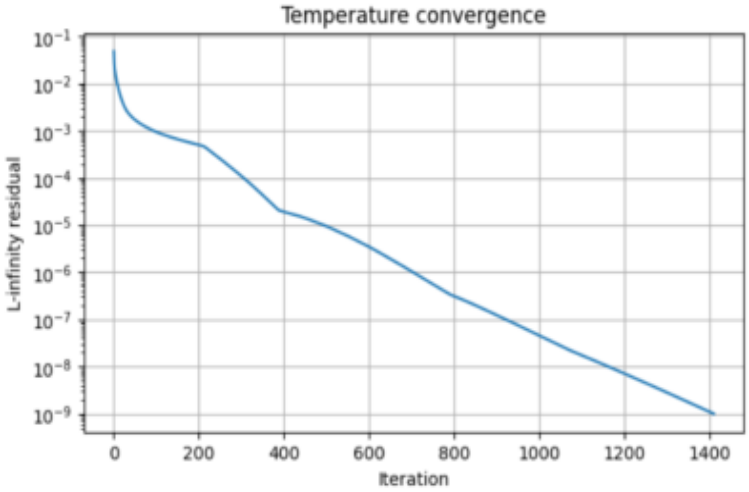


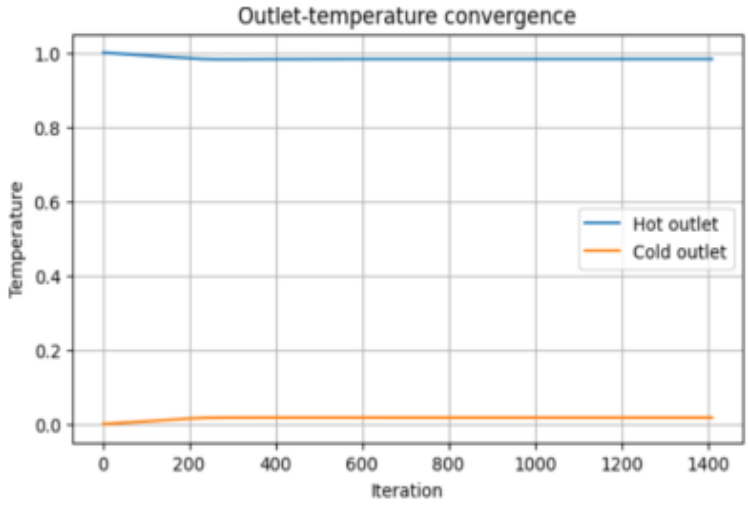
Original simulation output (source page 4)



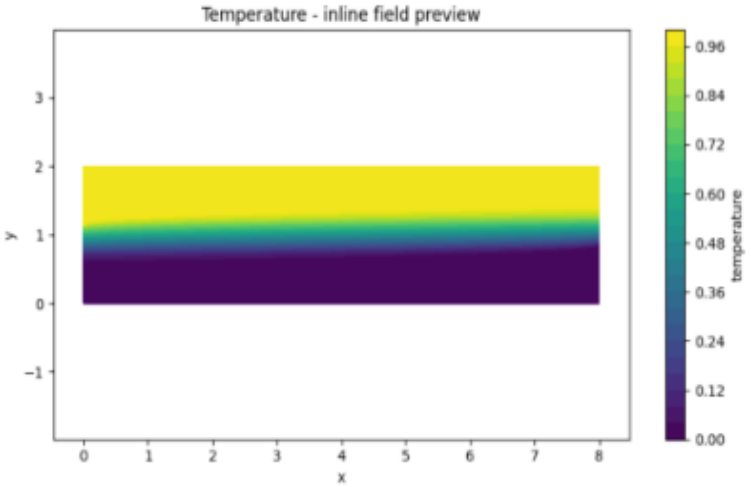
Original simulation output (source page 5)

For Counter flow system Raynold's number 200 and wall to fluid ratio 1

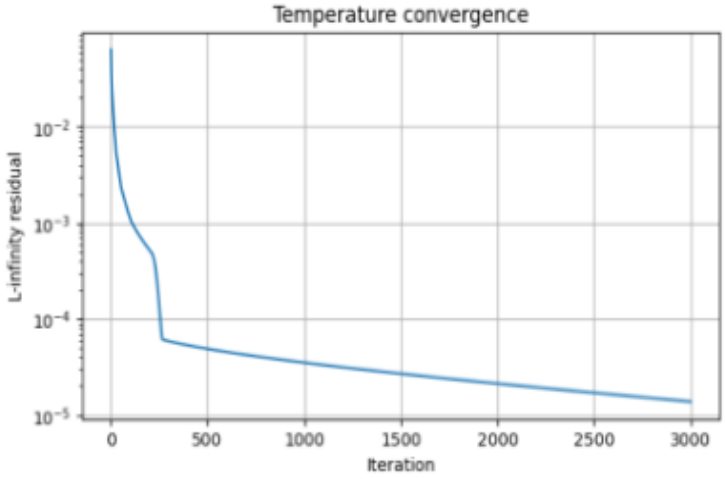




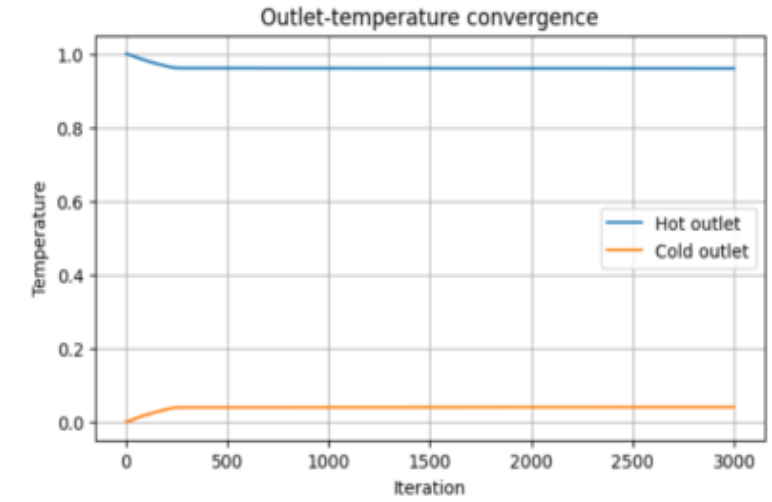
Original simulation output (source page 6)

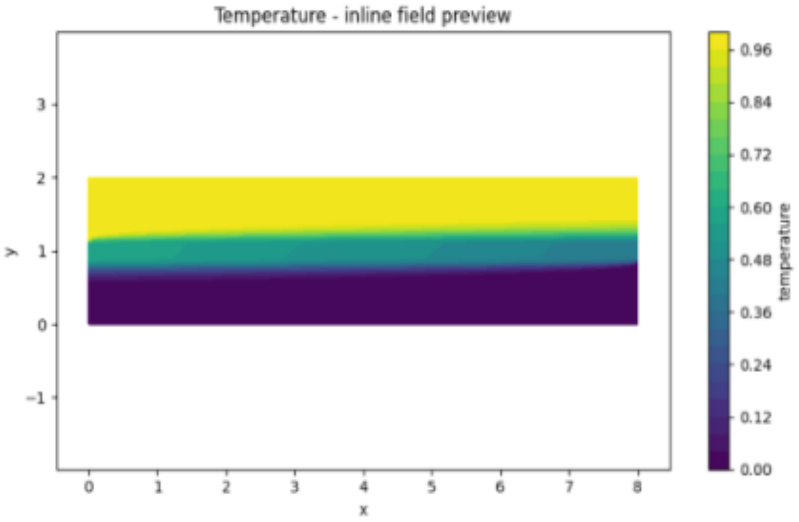


For Counter flow system Raynold's number 100 and wall to fluid ratio 100



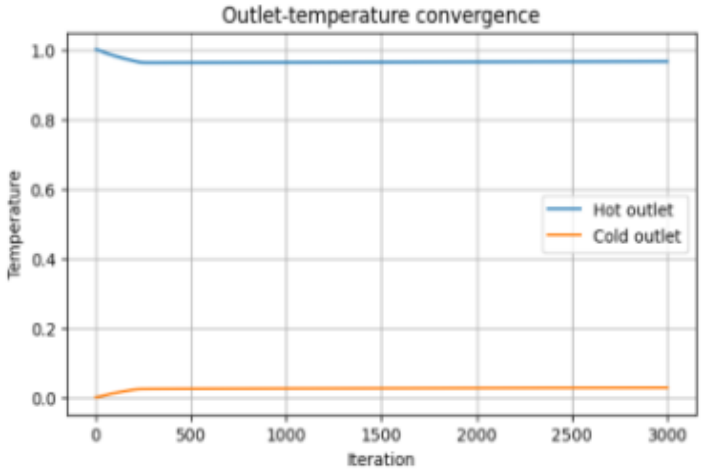
Original simulation output (source page 7)

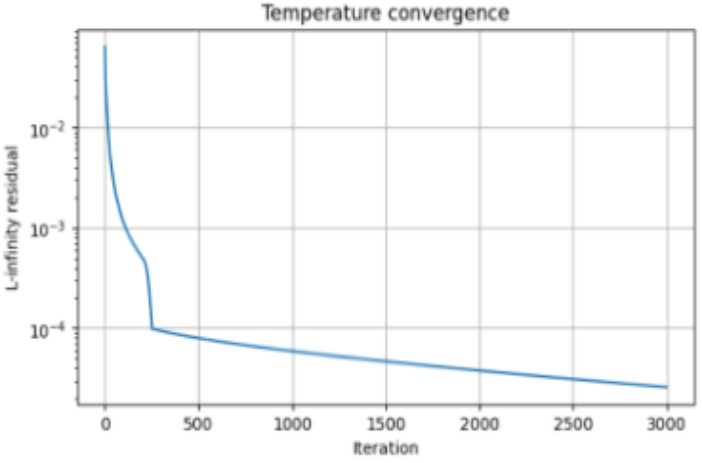




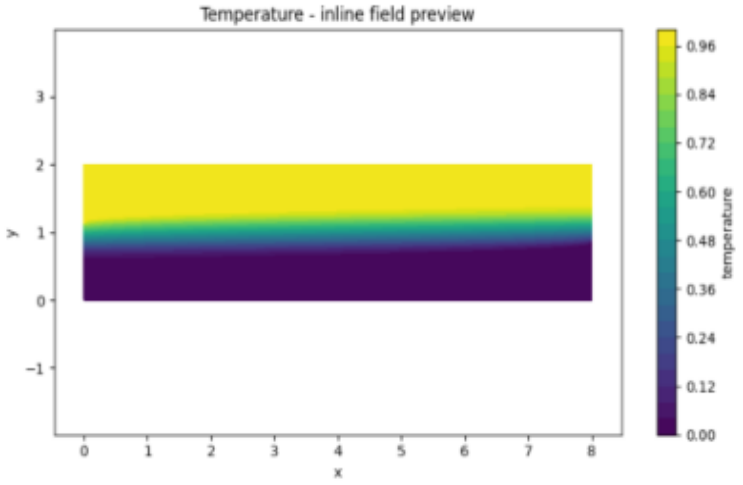
Original simulation output (source page 8)

For Counter flow system Raynold's number 200 and wall to fluid ratio 100





Original simulation output (source page 9)



Results:

Reynold's number for cold flow	Ratio of heat transfer coefficient	Hot inlet temperature	Hot Outlet temperature	Cold Outlet temp	Cold Outlet temperature
50	1	1	0.98	0	0.025
100	1	1	0.97	0	0.035
200	1	1	0.98	0	0.020
100	100	1	0.96	0	0.04
200	100	1	0.965	0	0.03

Conclusion:

The hot-stream inlet temperature was maintained at $T_{h,in}$ = 1.0, while the cold-stream inlet temperature was maintained at $Tc_{,in}$ = 0.0. For a wall-to-fluid ratio of 1, increasing the cold-stream Reynolds number from 50 to 100 increases the cold outlet temperature from 0.025 to 0.035, while the hot outlet temperature decreases from 0.98 to 0.97. At $Re_{(c)}$ = 200, the outlet temperatures are approximately 0.98 and 0.020 for the hot and cold streams, respectively. Increasing the wall-to-fluid ratio from 1 to 100 results in enhanced thermal interaction, with the hot outlet temperature decreasing to 0.96 at $Re_{(c)}$ = 100 and 0.965 at $Re_{(c)}$ = 200, while the corresponding cold outlet temperatures increase to 0.040 and 0.030, respectively.